

Chapter 1

The Scope and Method of Economics

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Economy versus Economics

Economy versus Economics

- Etymology of *economy*
 - Western point of view: Greek word *oikonomos*, which means "one who manages a household."
 - According to this point of view, economy means to manage a household.
 - Eastern point of view: Chinese word 經世濟民, which means "to rule the world and save the people."
- An economy is a complex *system* of production, consumption, and exchange activities that ultimately determine the *allocation* of resources.
- An economy can encompass a nation, a region, a single industry, or even just one family.
- Economics is the academic discipline that studies the principles and theories behind economic activities.

Definition and Scope of Economics

- Economics: The study of how individuals and societies choose to use the scarce given resources.
- Two major divisions of economics
 - Microeconomics: The branch of economics that examines the functioning of individual industries and the behavior of individual decision-making-units—that is, firms and households.
 - Macroeconomics: The branch of economics that examines the economic behavior of aggregates—income, employment, output, and so on—on a national scale.

Microeconomics & Macroeconomics

TABLE 1.1 Examples of Microeconomic and Macroeconomic Concerns

Division of Economics	Production	Prices	Income	Employment
Microeconomics	<i>Production/output in individual industries and businesses</i> How much steel How much office space How many cars	<i>Prices of individual goods and services</i> Price of medical care Price of gasoline Food prices Apartment rents	<i>Distribution of income and wealth</i> Wages in the auto industry Minimum wage Executive salaries Poverty	<i>Employment by individual businesses and industries</i> Jobs in the steel industry Number of employees in a firm Number of accountants
Macroeconomics	<i>National production/output</i> Total industrial output Gross domestic product Growth of output	<i>Aggregate price level</i> Consumer prices Producer prices Rate of inflation	<i>National income</i> Total wages and salaries Total corporate profits	<i>Employment and unemployment in the economy</i> Total number of jobs Unemployment rate

The Diverse Fields of Economics

TABLE 1.2 The Fields of Economics

<i>Behavioral economics</i>	Do aggregate household savings increase when we automatically enroll people in savings programs and let them opt out as opposed to requiring them to sign up?
<i>Comparative economic systems</i>	How does the resource allocation process differ in market versus command and control systems?
<i>Econometrics</i>	What inferences can we make based on conditional moment inequalities?
<i>Economic development</i>	Does increasing employment opportunities for girls in developing nations increase their educational achievement?
<i>Economic history</i>	How did the growth of railroads and improvement in transportation more generally change the U.S. banking systems in the nineteenth century?
<i>Environmental economics</i>	What effect would a tax on carbon have on emissions? Is a tax better or worse than rules?
<i>Finance</i>	Is high frequency trading socially beneficial?
<i>Health economics</i>	Do co-pays by patients change the choice and use of medicines by insured patients?
<i>The history of economic thought</i>	How did Aristotle think about just prices?
<i>Industrial organization</i>	How do we explain price wars in the airline industry?
<i>International economics</i>	What are the benefits and costs of free trade? Does concern about the environment change our views of free trade?
<i>Labor economics</i>	Will increasing the minimum wage decrease employment opportunities?
<i>Law and economics</i>	Does the current U.S. patent law increase or decrease the rate of innovation?
<i>Public economics</i>	Why is corruption more widespread in some countries than in others?
<i>Urban and regional economics</i>	Do enterprise zones improve employment opportunities in central cities?

Why Study Economics?

- Why did you *choose* to major in economics?
Why did you *choose* to take this course?

Why Study Economics?

- Why did you *choose* to major in economics?
Why did you *choose* to take this course?
- Three main reasons to study economics:
 - ① To learn a way of thinking
 - ② To understand society
 - ③ To be an informed citizen

To Learn a Way of Thinking

Three fundamental concepts

The study of economics teaches us a way of thinking and helps us make decisions.

- ➊ Opportunity Cost: The best alternative that we forgo or give up, when we make a choice or a decision.
- ➋ Marginalism: The process of analyzing the additional or incremental costs or benefits arising from a choice or decision.
- ➌ Efficient Markets–No Free Lunch: A market in which profit opportunities are eliminated almost instantaneously.

To Understand Society & To Be an Informed Citizen

- The study of economics is an essential part of the study of society.
- Societal changes are often driven by economics.
- To be an informed citizen requires a basic understanding of economics.
- Recommendable resources helping us to stay informed
 - News article: The Economist, The New York Times, The Wall Street Journal, Bloomberg, and
 - Data: WEO (World Economic Outlook), OECD Economic Outlook, World Bank, IMF, Bureau of Labor Statistics, and
 - YouTube clip: CNBC, Vox, and
- ★ Read skeptically and think critically!

How to Study Economics

Two Approaches

- Positive economics: An approach to economics that seeks to understand behavior and the operation of systems without making judgements. It describes what exists and how it works.
 - Theories and models are tools used to answer positive questions in economics.
- Normative economics (policy economics): An approach to economics that analyzes outcomes of economic behavior, evaluates them as good or bad, and may prescribe courses of action.
 - Efficiency, equity, growth, and stability are the four criteria frequently applied in judging economic outcomes.

Theories and Models

Positive Approach

- A model is a simplified version of reality.
 - Human body → plastic replica
 - City/state/nation/continent → map
 - ★ Economy → formal statement of a theory (words, graphs, and mathematical equations)
- Cautions and pitfalls when setting up a model
 - Note: Problem of causality
 - All else equal (ceteris paribus): discarding confounding factors
 - Distinction between causality and correlation
 - Post hoc, ergo propter hoc fallacy
 - Spurious relation
 - ★ Does your roommate matter for your grades?
- Testing theories and models: empirical economics

Economic Policy

Normative Approach

- Four criteria in evaluating policies or economic outcomes.
 - ① Efficiency: no waste principle.

Is the current allocation of resources the best? If there exist some inefficiency in economy, how (how much) can (should) we reduce it?
 - ② Equity: fairness principle.

Is the current resource allocation fair?
 - ③ Growth: economic growth and standard of living
Does the policy encourage/hamper economic growth?
 - ④ Stability: low inflation and full employment of resources

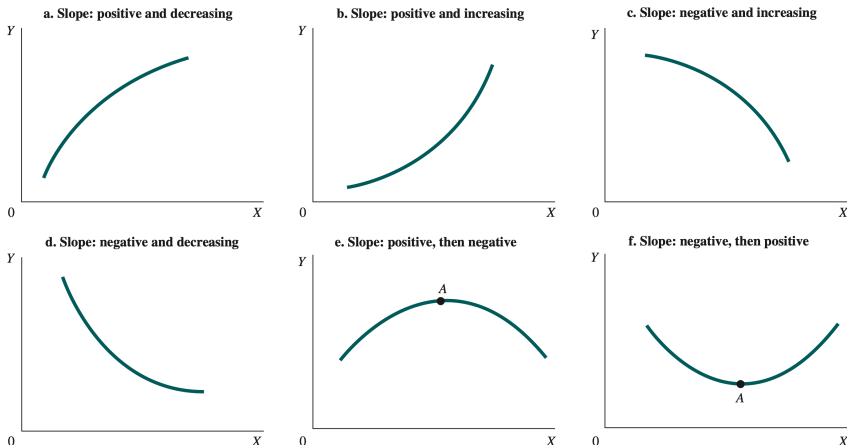
How to Read and Understand Graphs

- A graph is a two-dimensional representation of a set of variables (numbers, observations, data).
 - Elements: X-axis, Y-axis, origin, X-intercept, Y-intercept
 - Slope: positive/zero/negative, changing/constant

$$\text{slope} = \frac{\Delta Y}{\Delta X} = \frac{Y_2 - Y_1}{X_2 - X_1}.$$

Note: movement *along* the curve vs. movement *of* the curve

How to Read and Understand Graphs



▲ **FIGURE 1A.4** Changing Slopes Along Curves

MyLab Economics Concept Check